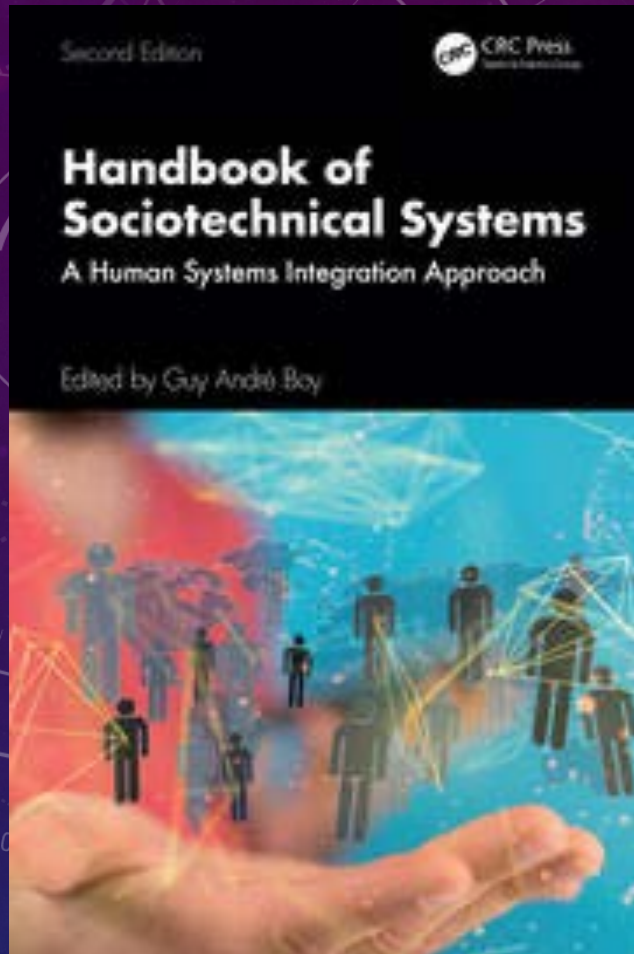




SOCIOTECHNICAL SYSTEMS: AN HSI APPROACH

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AIR & SPACE ACADEMY
INTERNATIONAL ACADEMY OF ASTRONAUTICS
INCOSE FELLOW

EVOLUTION FROM AUTOMATION ISSUES TO TANGIBILITY ISSUES

This lecture will focus on demonstrating how human factors and organizations play a crucial role in the design, development, and evaluation of sociotechnical systems, such as those found in aeronautics, nuclear engineering, space, hospital medicine, and railways, among others.

We will discuss systems of systems, functions and structures, roles, contexts, and resources.

We will also explore the crucial importance of the concept of emergence as we begin to study and design complex systems that are also critical to safety.

We will address risk-taking, contingency management, flexibility, human skills and knowledge, as well as the organizational structures needed to solve problems, particularly in crisis situations.

All of this will be discussed in the current and future context of digital engineering, which will highlight issues of physical and figurative tangibility.

<https://www.routledge.com/Handbook-of-Sociotechnical-Systems-A-Human-Systems-Integration-Approach/AndreBoy/p/book/9781041122289>

**“NO HUMAN-CENTERED
DESIGN WILL BE REALLY
POSSIBLE WITHOUT AN
APPROPRIATE
ORGANIZATION”**



GROUPTHINK

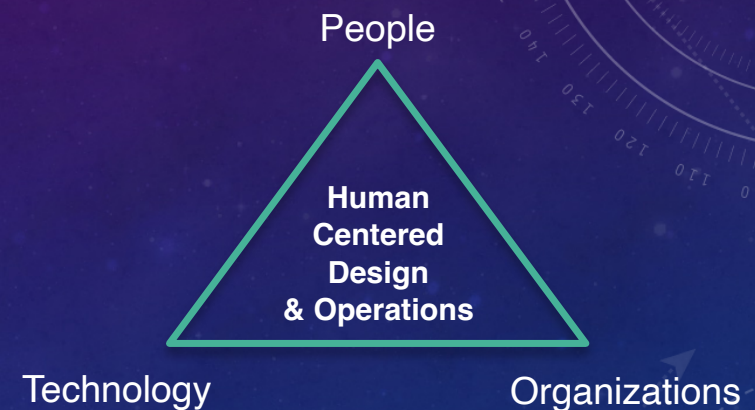


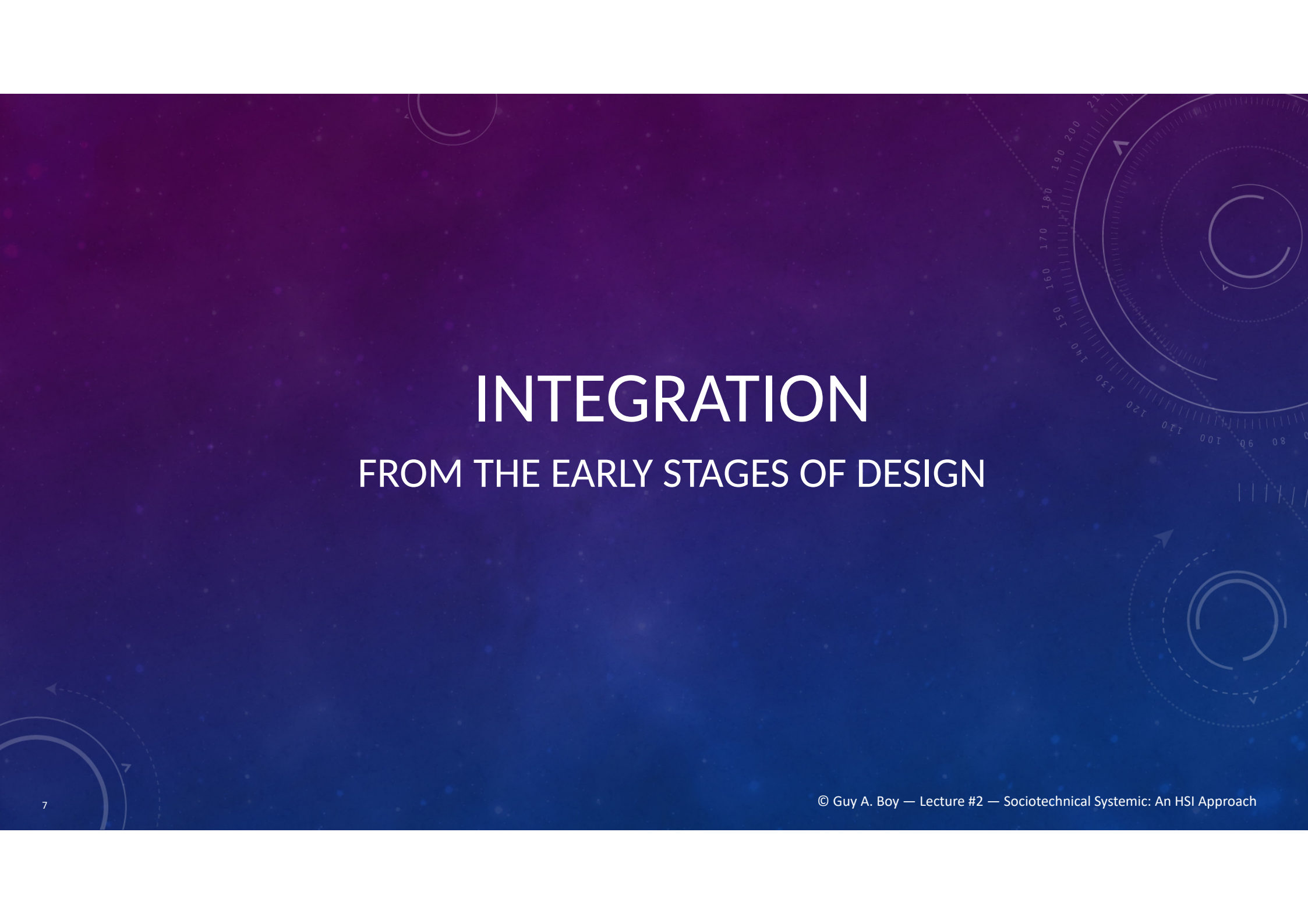
OUTLINE OF TODAY'S COURSE

- Organization Design & Management?
- The crucial issue of Integration
- Bringing an engineering system to maturity
- Business models
- Solutions for the future...

WHY ODM IS SO IMPORTANT?

- Technology, Organizations and People
 - Evolution of new technology
 - Work force demographics change
 - Attitudes and values change
 - Real quality of work life
 - Structures and processes.
- Productivity and efficiency of work systems
- Macro-ergonomics methods and applications:
 - Organizational and job design, change management
 - Socio-technical systems



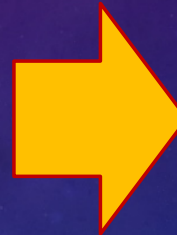
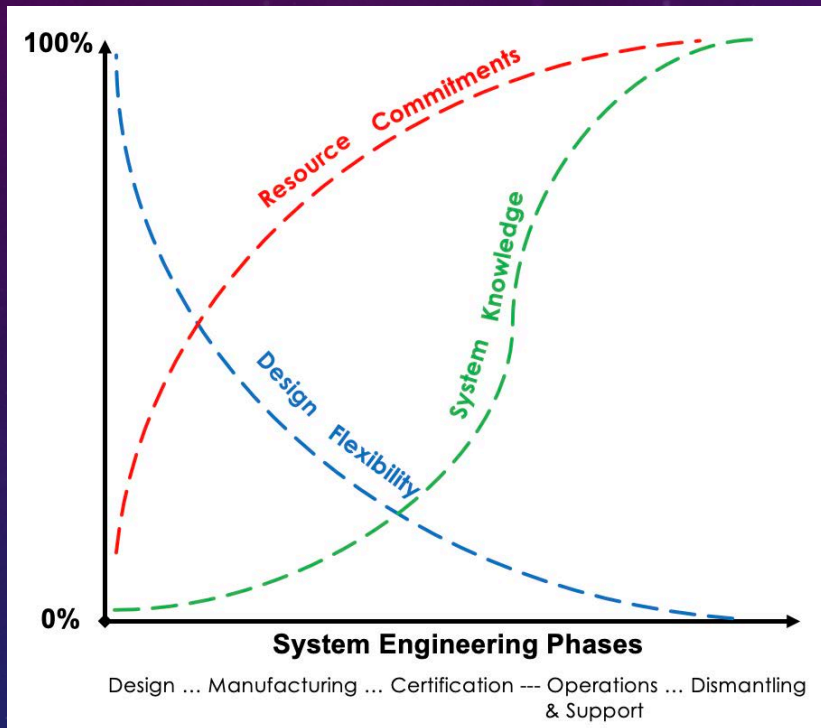


INTEGRATION

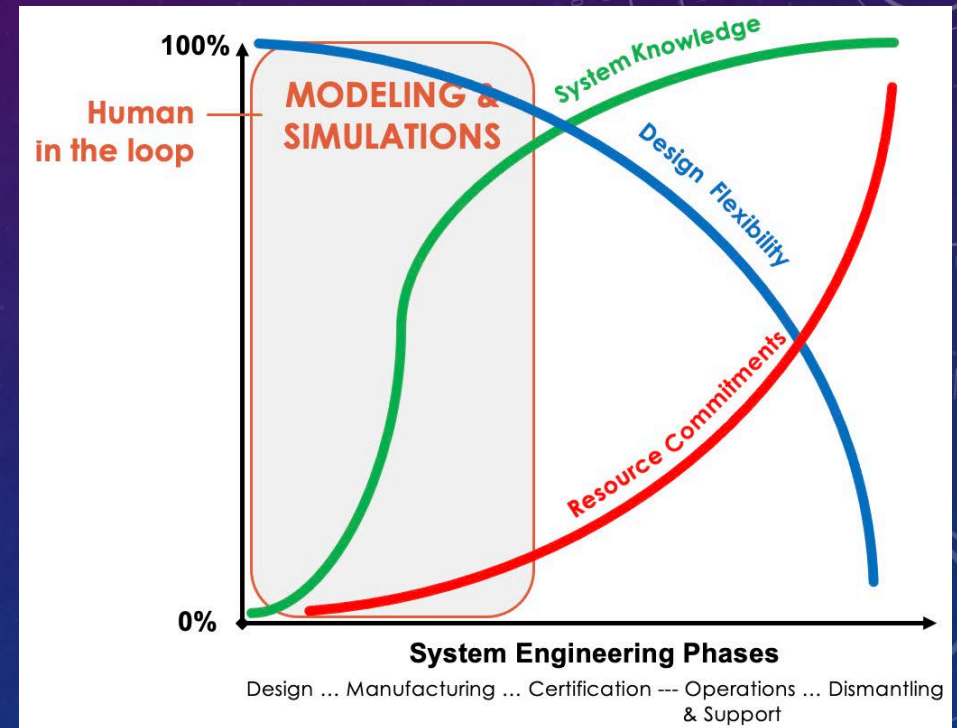
FROM THE EARLY STAGES OF DESIGN

LIFE-CYCLED HUMAN SYSTEMS INTEGRATION

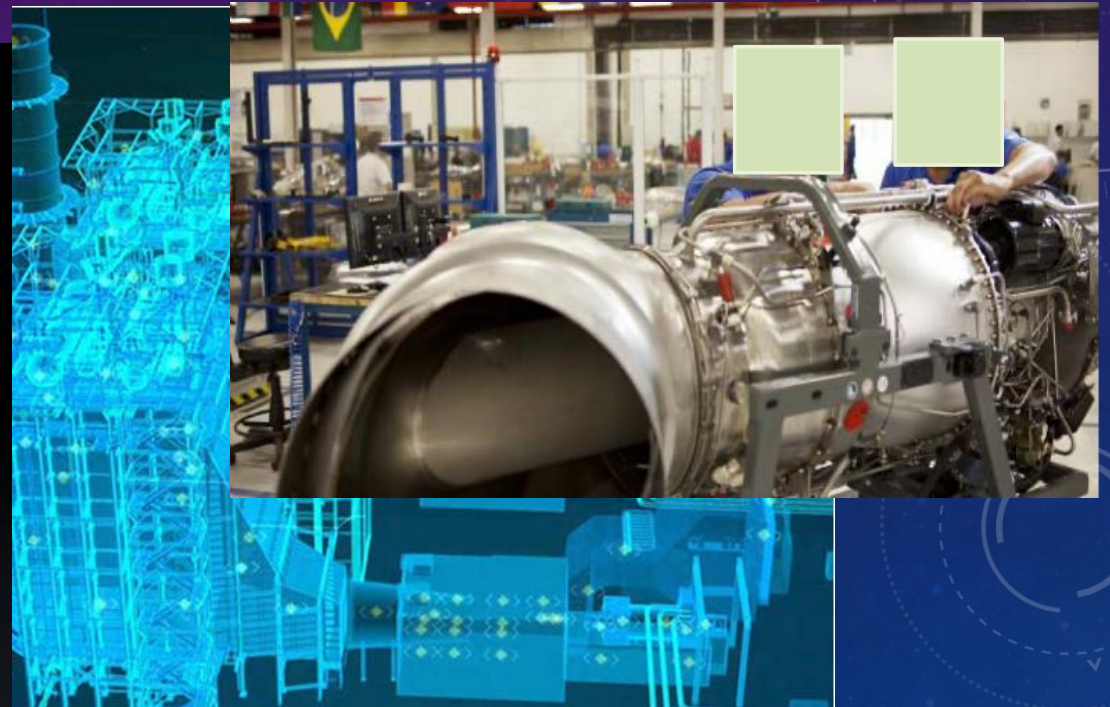
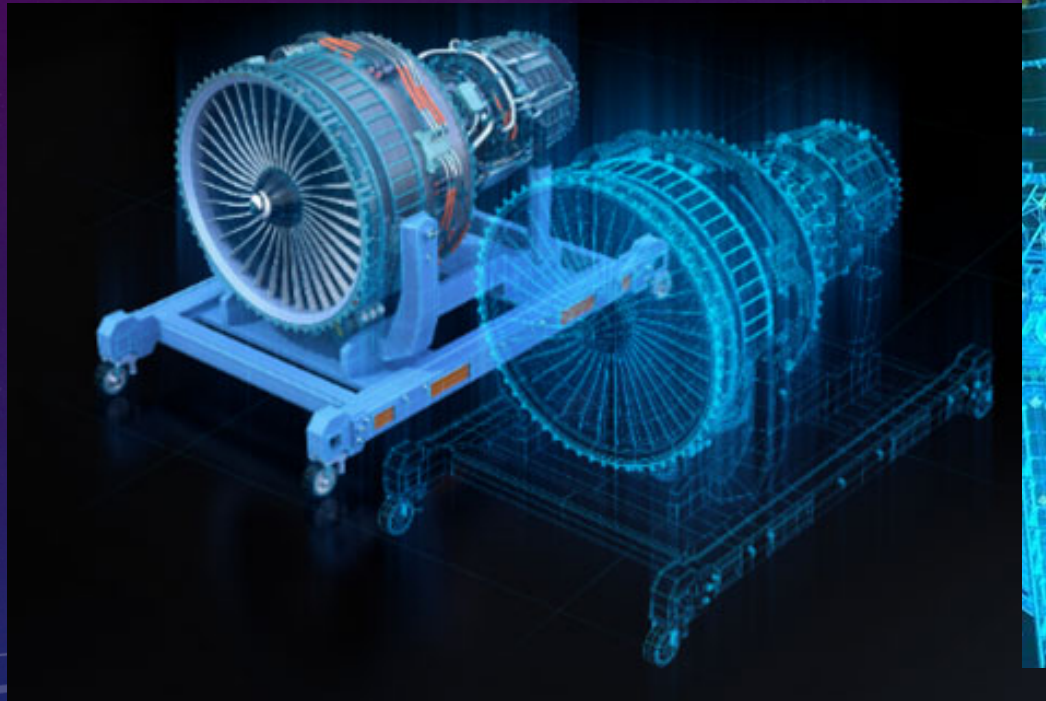
Technology-centered



Human-centered

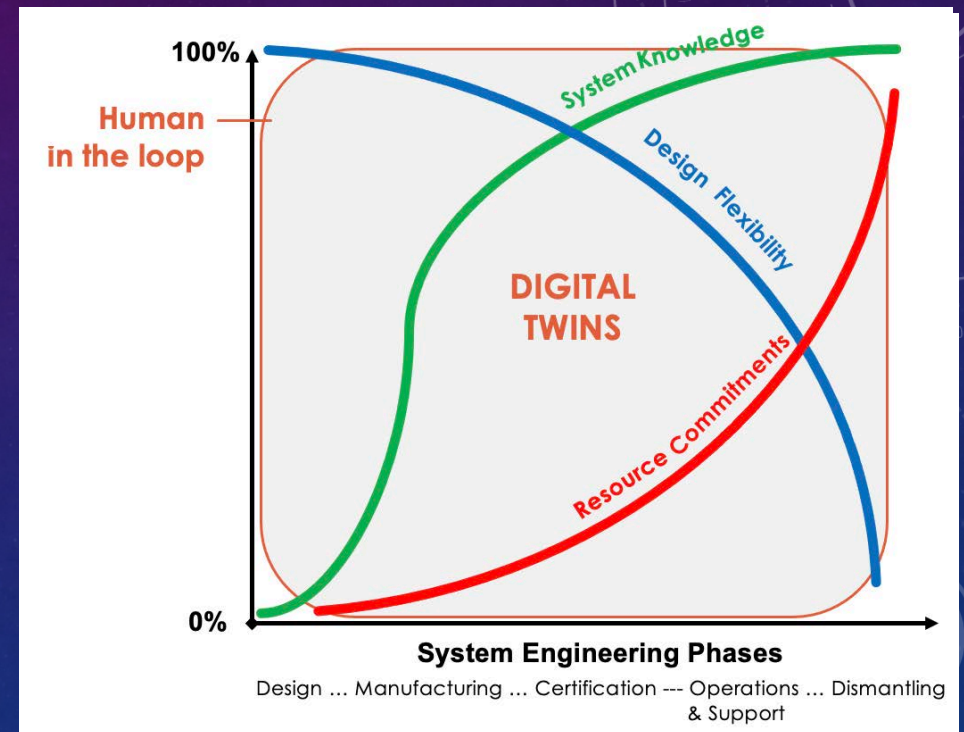


HUMAN-CENTERED DESIGN OF A DIGITAL TWIN FOR HELICOPTER ENGINE MAINTENANCE

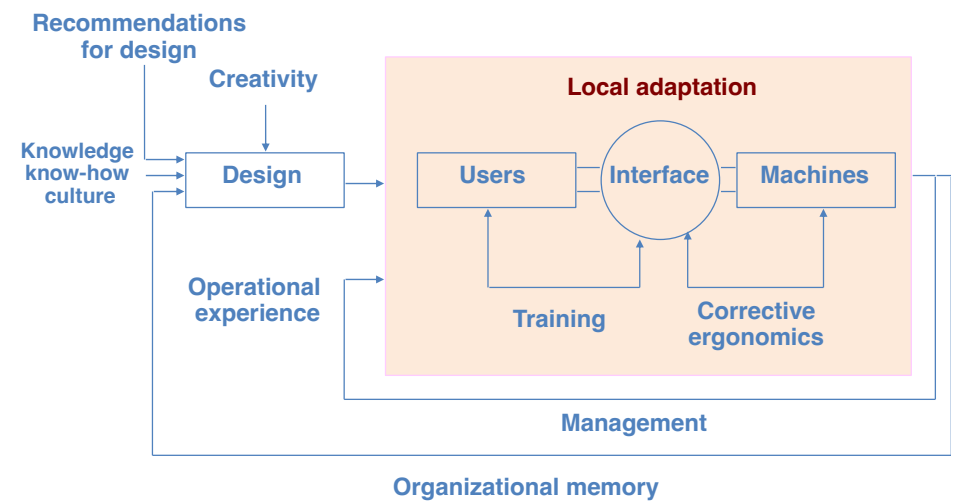


DIGITAL TWINS

- Expanding HITLS
 - During the whole life cycle
 - “what if?”
- Vivid documentation
 - Integration of experience feedback
 - Organizational memory
- Digital twins as virtual assistants
 - Multi-agent collaboration
 - Mediators for collaborative work

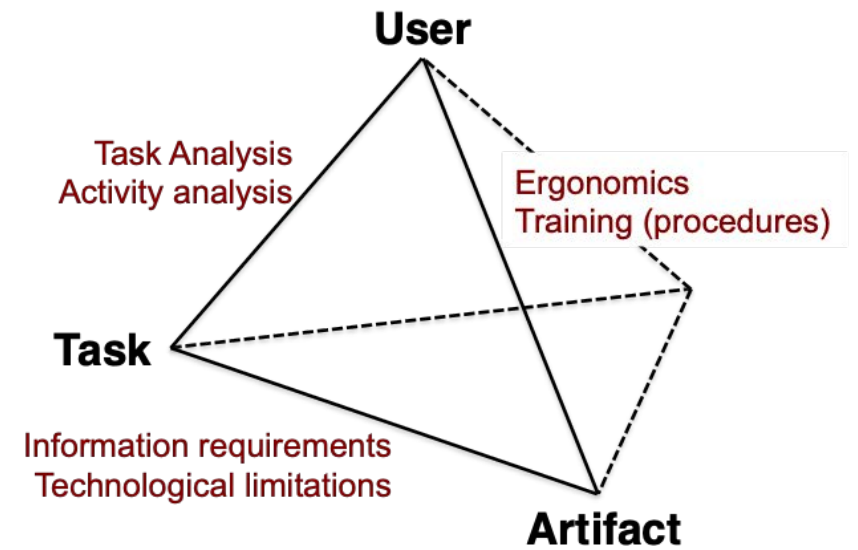


GLOBAL ERGONOMICS



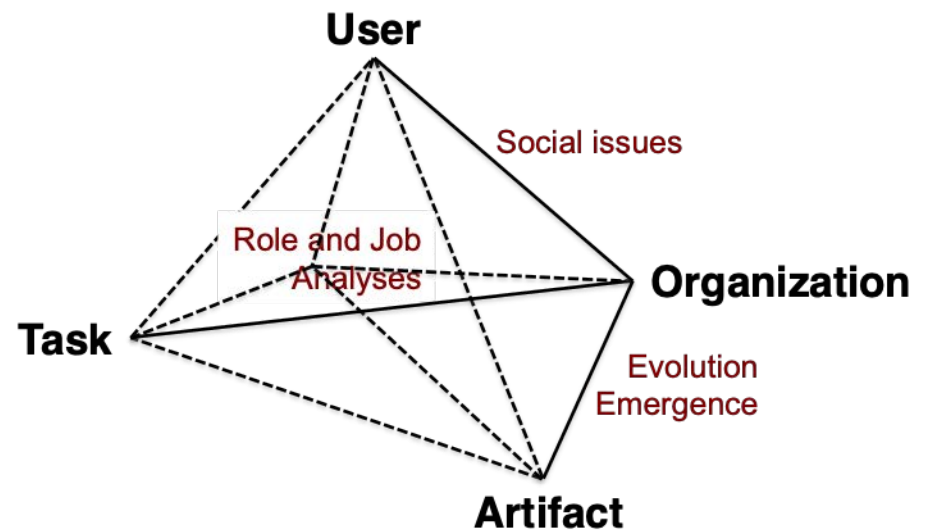
AUTOS...

THE AUT TRIANGLE



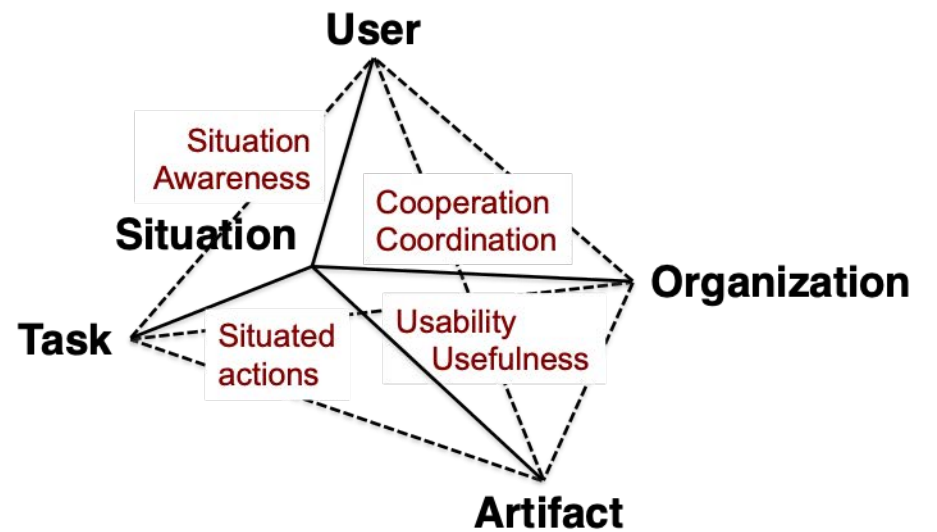
AUTOS...

THE AUTO TETRAHEDRON



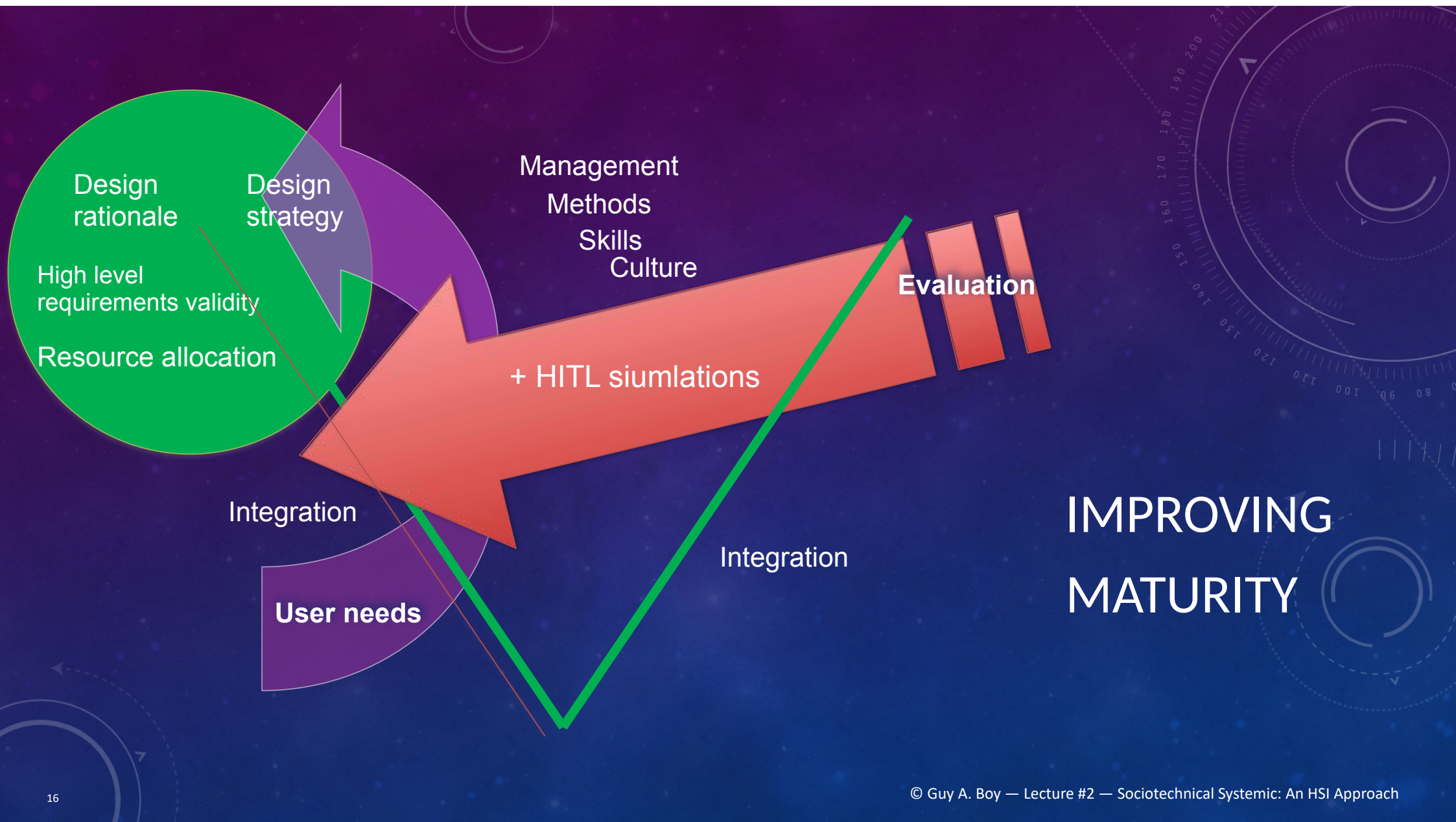
AUTOS... AGAIN!

THE AUTOS PYRAMID



A380 FLIGHT TESTS & CERTIFICATION

- Approval and Authorization of:
 - Operations
 - Procedures
 - People
 - Processes
 - Aircraft
 - Facilities
 - Equipment



- Can we predict the future?
No!
- But, we can make forecasts...
Abduct possible alternatives
Choose the best alternatives



DEALING WITH UNCERTAINTY

Educated Common Sense



Red sky at night, shepherd's delight



CONCRETE vs. ABSTRACT

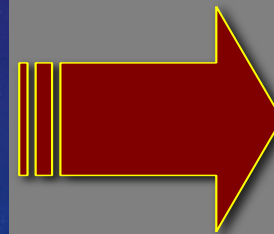
Physical Tangibility vs. Figurative Tangibility



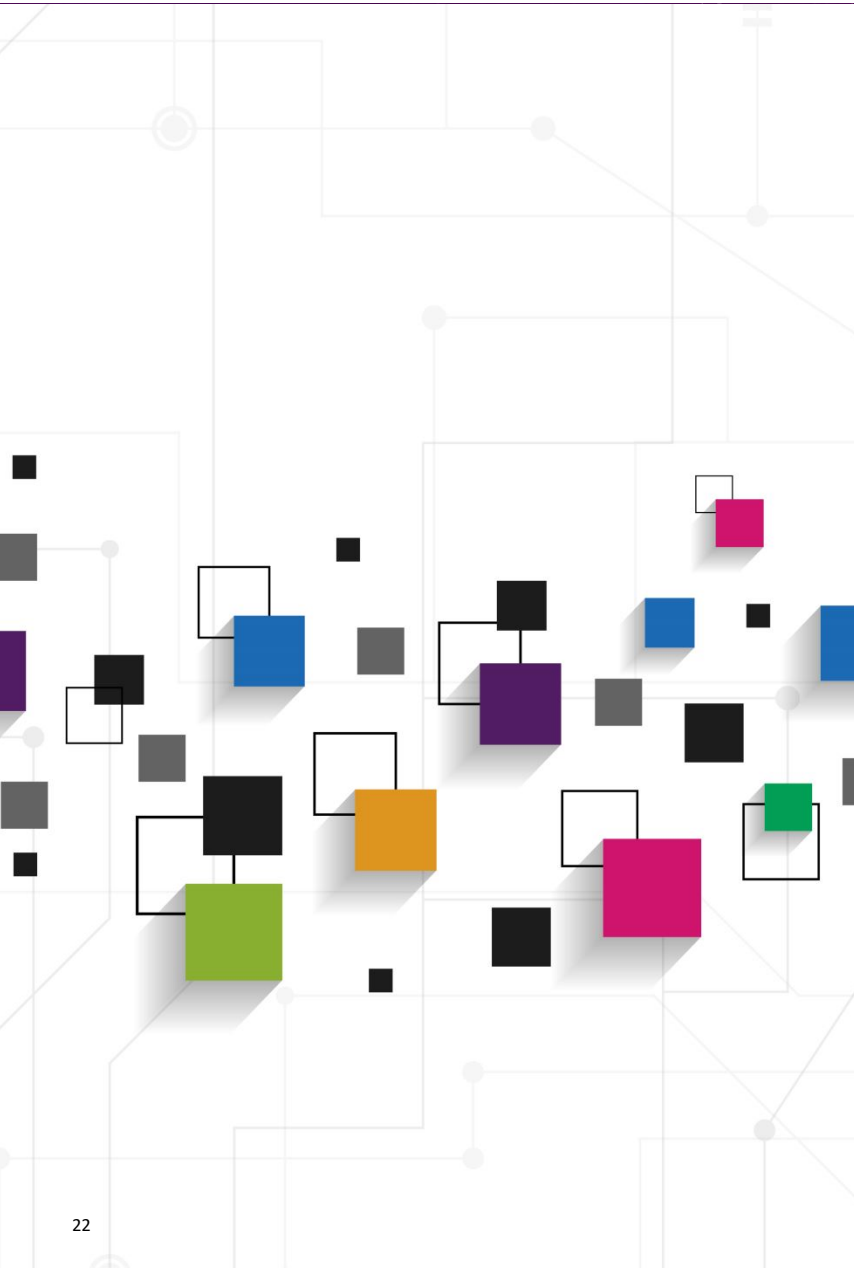
ABDUCTION

$a \rightarrow b$

- Allows inferring a as an explanation of b
- A form of intuitive reasoning that consists in removing improbable solutions



Intuition
Pragmatism
Experience
Competence



THE CRUCIAL ISSUE OF INTEGRATION

- Bringing together component subsystems into one system and ensuring that the subsystems function together as a system
- Examples of integration techniques:
 - Computer networking
 - Enterprise application integration
 - Business process management
 - Manual programming
 - ...

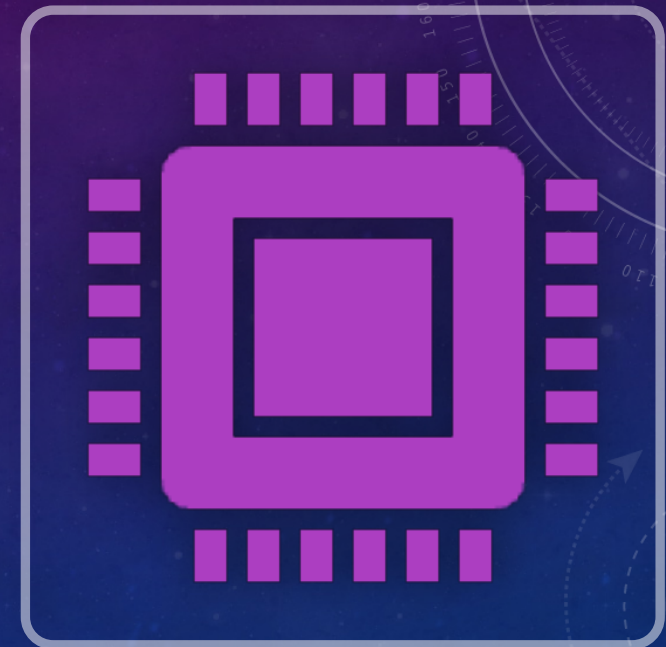
ENTERPRISE INTEGRATION NEEDS

- Identify the right information
 - *Requires precise knowledge of the information needed and created by the different activities in the enterprise operation*
 - *Knowledge to be structured in the form of an accurate model of the enterprise operation*
 - *A model which describes product and administrative information, resources and organizational aspects of the operational processes and allows what-if analysis in order to optimize these processes*



ENTERPRISE INTEGRATION NEEDS

- Provide the right information at the right place
 - *Requires information sharing systems and integration platforms capable of handling information transaction across heterogeneous environments*
 - *Environments that consists of heterogeneous hardware, different operating systems and monolithic software applications (legacy systems)*
 - *Environments that cross organizational boundaries and link the operation of different organizations on a temporal basis and with short set-up times and limited time horizon (extended and virtual enterprises)*



ENTERPRISE INTEGRATION NEEDS

- Update the information in real time to reflect the actual state of the enterprise operation
 - *Requires not only the update of the operational data (information created during the operation, but adapting to environmental changes as well*
 - *Changes that may originate from new customer demands, new technology, new legislation or new philosophies of the society at large*
 - *Changes that may require modification of the operational processes, the human organization or even the overall scope and goals of the enterprise*



ENTERPRISE INTEGRATION NEEDS

- Co-ordinate business processes
 - *Requires precise modeling of the enterprise operation in terms of business processes, their relations with each other, with information, resources and organization*
 - *Goes far beyond exchange of information and information sharing*
 - *Considers decisional capabilities and know-how within the enterprise for real time decision support and evaluation of operational alternatives*



ENTERPRISE INTEGRATION NEEDS

- Organize and adapt the enterprise
 - *Requires very detailed and up-to-date knowledge of both the current state of the enterprise operation and its environment (market, technology, society)*
 - *Knowledge which has to be available a priori and very well structured to allow easy identification of and access to relevant information*



BRINGING PRODUCT TO MATURITY

- High level requirements are crucial
- Incremental testing is crucial
- Human-system integration is crucial
- Principles: safety, efficiency and comfort

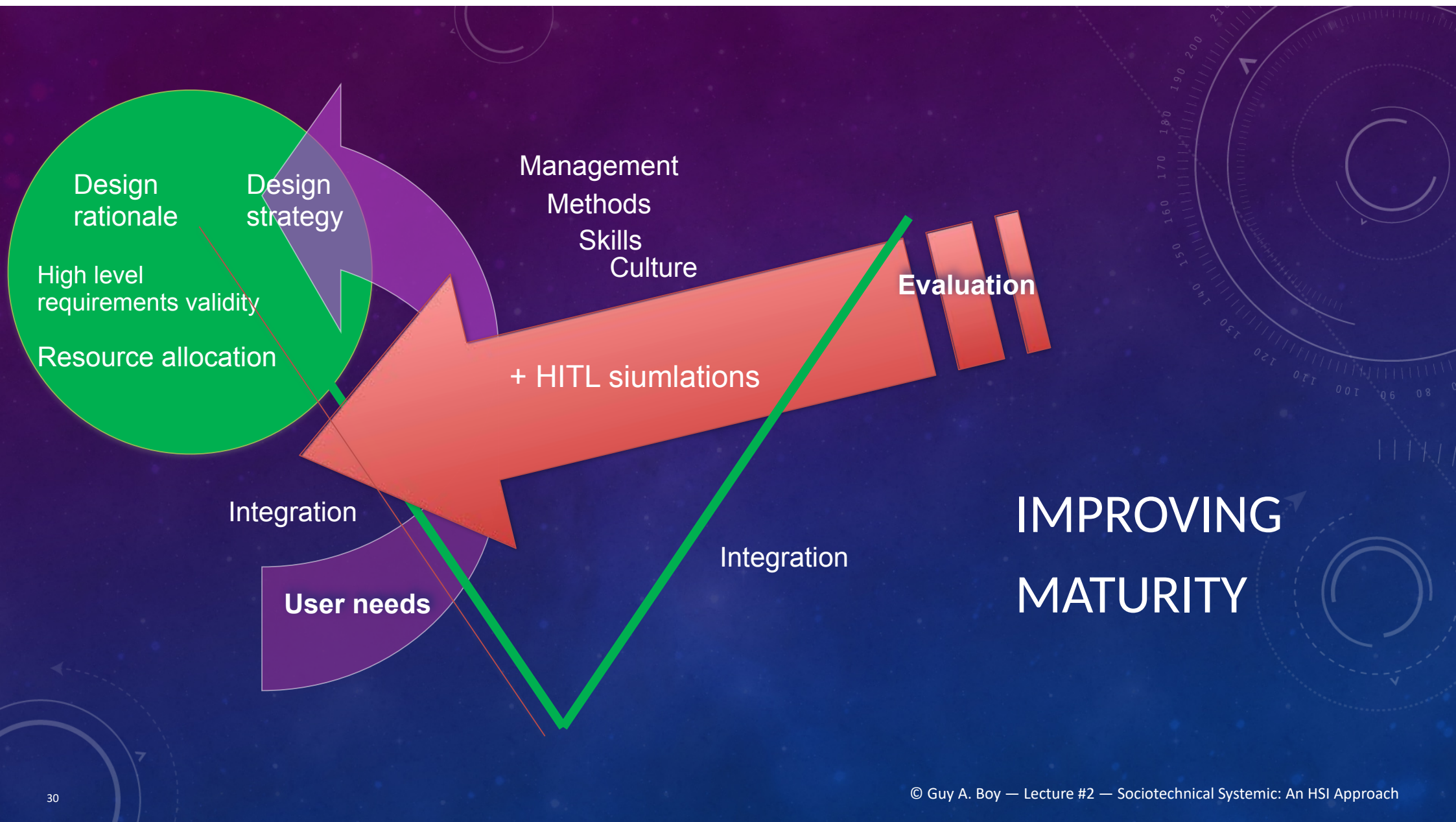


CMMI

- Capacity Maturity Model Integration
- Process improvement methodology
- Developed at CMU Software Engineering Institute (SEI)
 - <http://www.sei.cmu.edu/cmmi/>
- Used by many company around the world
- Does not seem to work! why?

But there is more to be considered



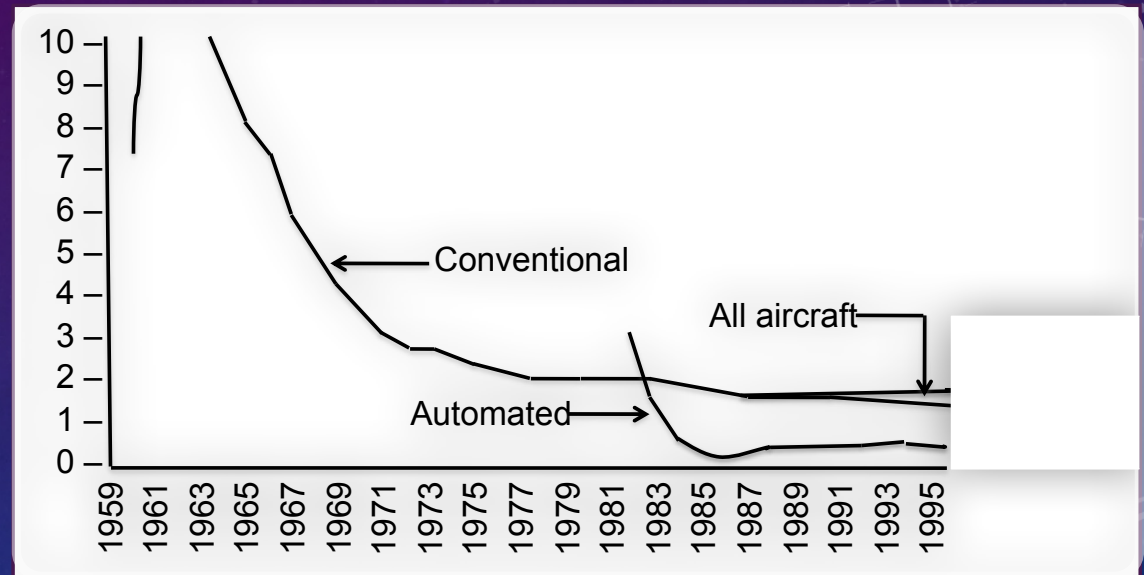




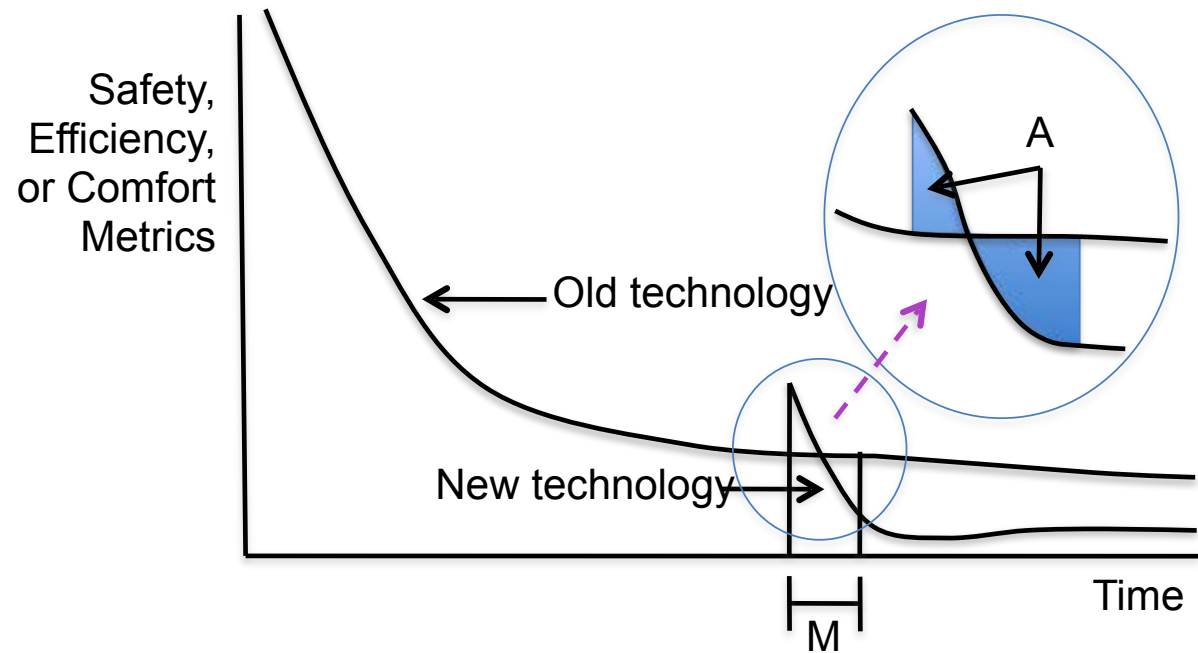
HUMAN-IN-THE-LOOP SIMULATIONS

SAFETY MATURITY CURVES

- Evolution of hull losses for conventional cockpits and glass cockpits (automated)
- per million of departures, presented by Airbus Industrie at the Flight Safety Conf. in 1997



SAFETY MATURITY CURVES



M: Maturity period
A: Acceptability area



THE INTERACTIVE COCKPIT!

WHAT REALLY MATTERS HERE?

- Technology
- Organization
- People



Motivation

Involvement

Competence

Evolution

MOTIVATION AT WORK

3 factors lead to better performance and personal satisfaction

- **Autonomy**
 - Desire to be self-directed
 - If you want **Engagement**, self-directed is better
- **Mastery**
 - The urge to get better at stuff (during free time)
 - Have fun and volunteer your time (for free!)
 - Challenge mastery and make a contribution
- **Purpose**
 - Transcendent purpose in the organization
 - When profit motives get unmoored from purpose motive, bad things happen (bad service) – money, money, money!!!



MOTIVATION THEORY

Maximizing Purpose

- *“Our goal is to be destructive but in the cause of making the world a better place”*
 - Cool!... No phone bills...
- *“I want to put a Ding in the Universe”*
 - Steve Jobs



WHAT REALLY MATERS HERE?

from purpose to means
and not
from means to purpose



© Guy A. Boy — Lecture #2 — Sociotechnical Systemic: An HSI Approach



“I AM NOT A HORSE,
I AM A PERSON...”

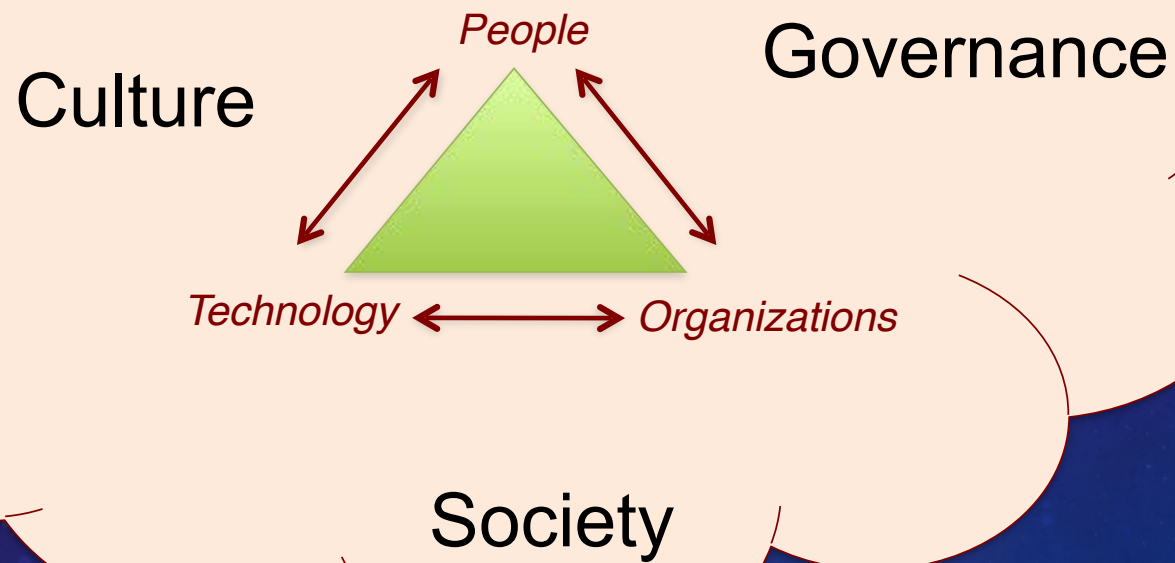
→ MAKE ORGANIZATIONS
THAT MAKES US BETTER
AND MAKE THE WORLD
A LITTLE BIT BETTER!

- It's time to get up-to-date
- with *governance design!*
 - Society is an invention, not a “natural given”.
 - Everything in society was created by humans, and needs to be re-created by us
- You need to *become a social inventor*, especially of governance systems
 - Think in terms of what “could” or “should be” not what “will be” or is “most likely”.
 - The future of governance cannot be predicted, but it can be invented

WORLD EVOLUTION

... from Purpose to Means

REMEMBER...



WORLD EVOLUTION...



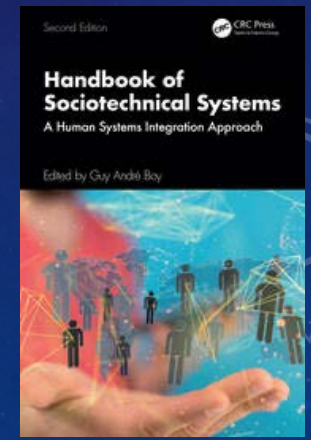
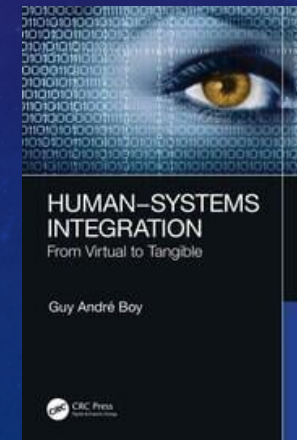
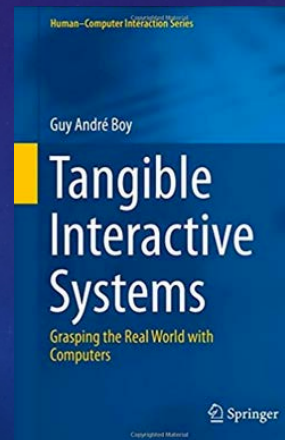
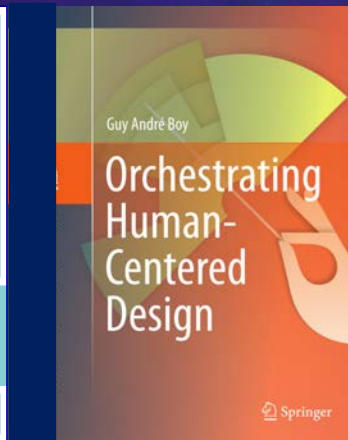
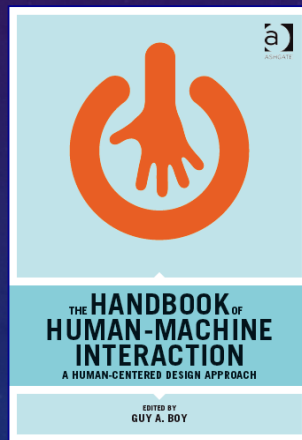
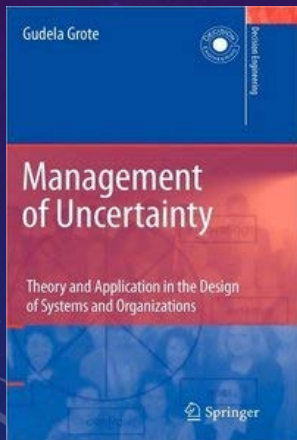
WORLD EVOLUTION...

- Who has the authority?
 - control and accountability

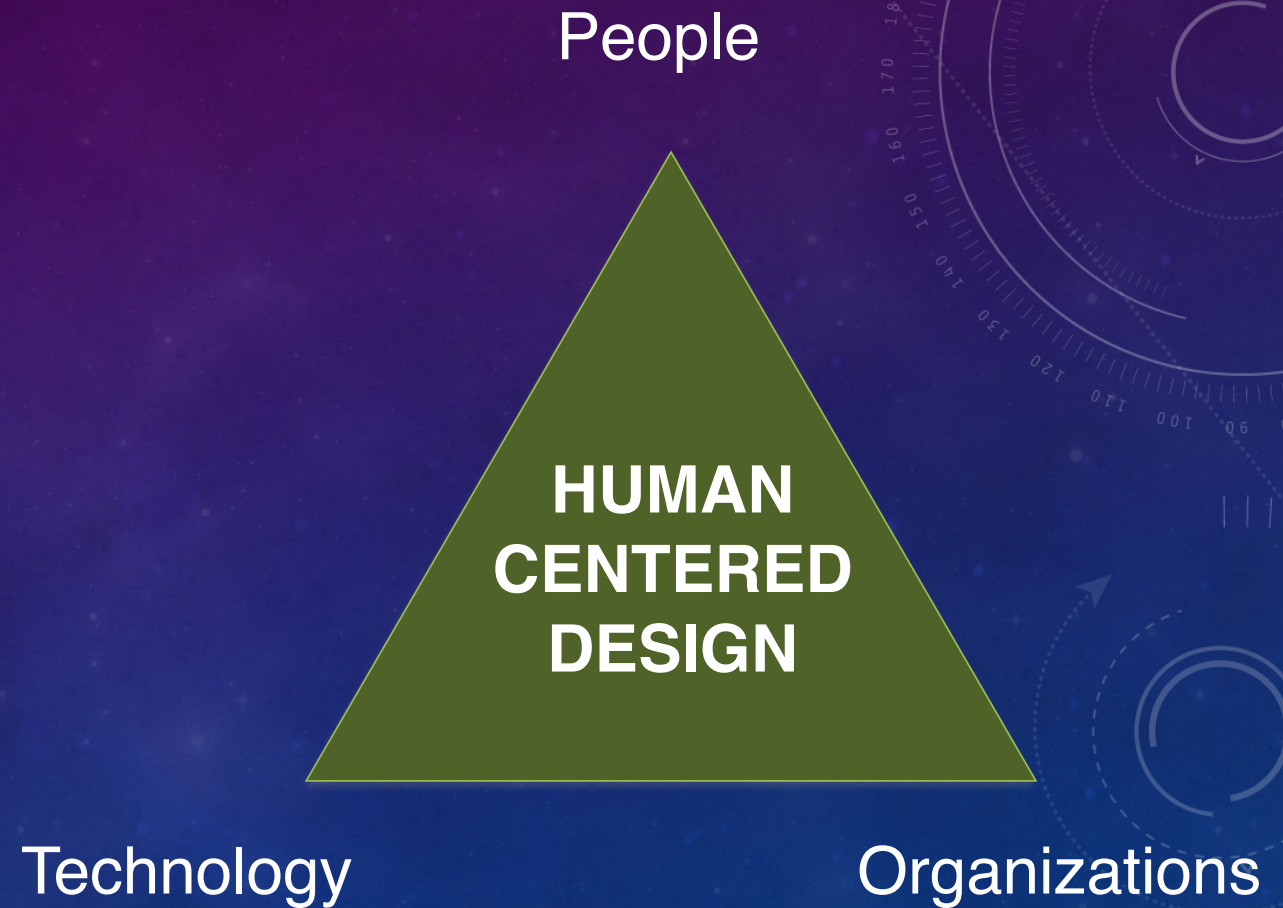


REFERENCES FOR THE COURSE

- Handouts
- My web site
 - <https://profils.estia.fr/g.boy>
- ... be curious!



Be TOP!

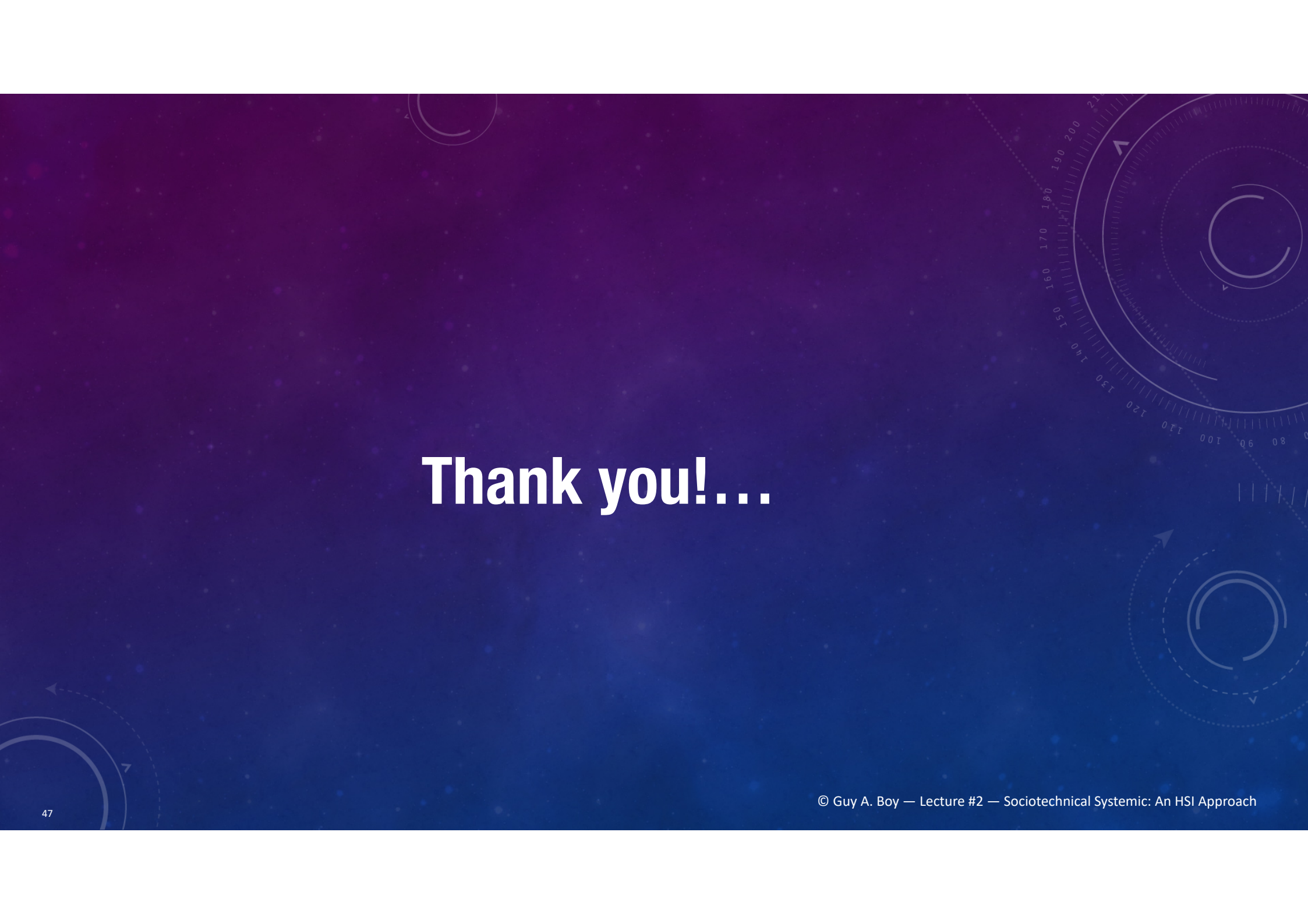


AUTONOMOUS VEHICLES...

Group Exercise



- Investigate trains, cars, and aircraft...
- What are the relevant concepts?
- What is the difference between autonomy and automation?
- How do you state the problem?
- What are the solutions?
- Imagine possible futures...



Thank you!...